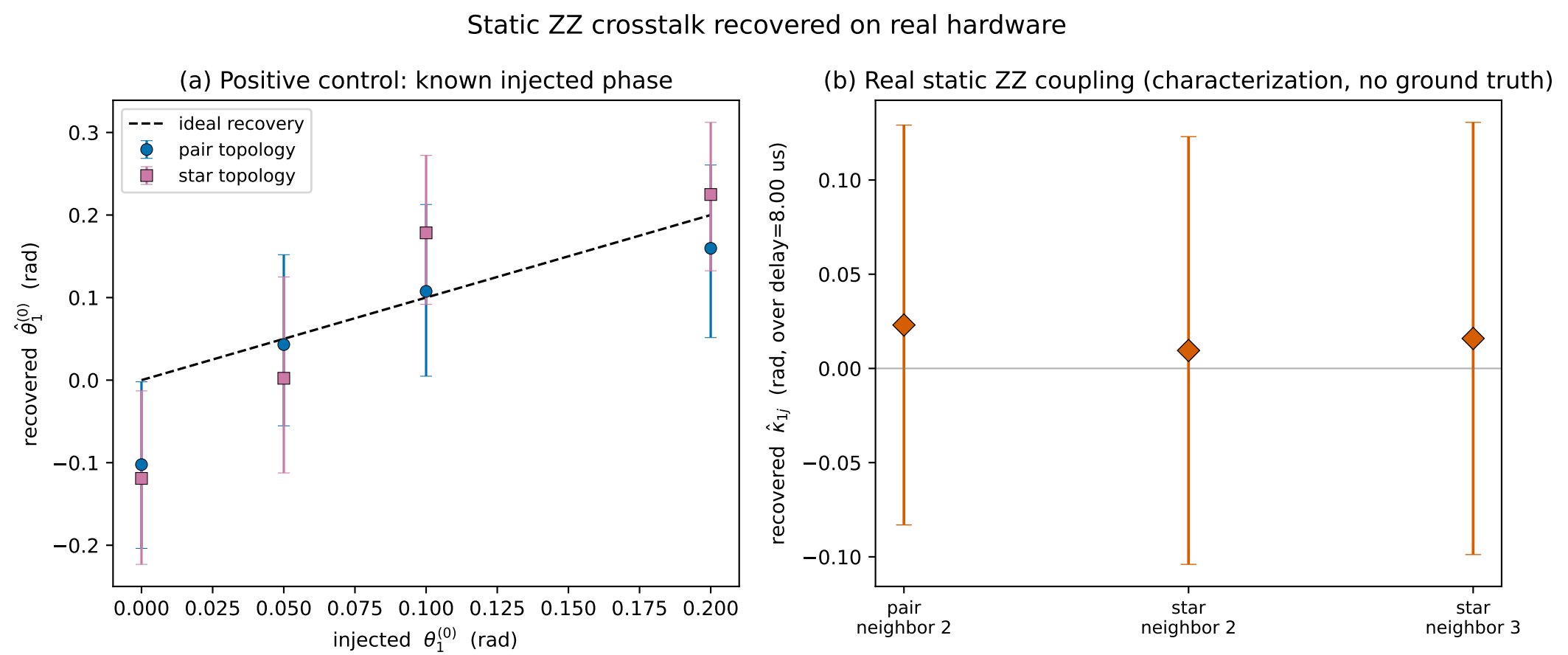
Static ZZ crosstalk recovered on real hardware
(a) Positive control: known injected phase
ideal recovery
pair topology
star topology
recovered $\hat{\theta}_1^{(0)}$ (rad)
injected $\theta_1^{(0)}$ (rad)
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.3
0.2
0.1
0.0
−0.1
−0.2
(b) Real static ZZ coupling (characterization, no ground truth)
recovered $\hat{\kappa}_{1j}$ (rad, over delay=8.00 us)
0.10
0.05
0.00
−0.05
−0.10
pair
neighbor 2
star
neighbor 2
star
neighbor 3
backend=ibm_marrakesh, qubits=[53, 54], date=2026-08-16, shots=2000/circuit, delay=8.00 us